UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

The X machine was the first automatic electronic digital computer that uses binary system. It has 30 arithmetic/logic unit that can add and subtract, a rotating drum memory that can held thirty binary numbers of fifty digits each and input/output components. Figure 1 shows the architecture of X machine.

bose-iO to
drum timing contot
drum
base -2
contacts
brushes
boost carry-
dYVhf
brushes
motor
switch
r box
gut
pwer suppty
regulator
crd
(7 duel triodes)
circuit

Figure 1: Machine X

a) What is X machine?
b) Name the inventor of X machine.
c) Based on Figure 1, identify the input and output components for X machine.

Source: CSC429 (7).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Parallel and multiprocessor architecture

Complete the following taxonomy of parallel processor architecture with appropriate example.

Single Instruction Multiple
Data Stream
ii
Processor Organization
Multiple Instruction Multiple
Data Stream
v

Source: CSC429 (5).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.

Instruction
Opcode
Address A

Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Data representation and number systems

Convert -20.125 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: -20.12510 = 0xC1A10000 (sign=1, exponent=10000011, fraction=01000010000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A5 (10 marks)

Topic: Memory hierarchy and cache

Describe FOUR (4) type of access methods based on hardware implementation of the storage device. Provide ONE (1) appropriate example for each access type.

Source: CSC429 (4).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Computer architecture and CPU organisation

a) Discuss the TWO (2) processes in the machine cycle that are governed by the main memory.
b) List TWO (2) operations that are comprised in the Execute phase.
c) Explain what happens in the Program Counter (PC) and Instruction Register (IR) during the Fetch cycle.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Input/output and interrupts

a) Describe THREE (3) types of interfaces to external devices.
b) Explain TWO (2) types of interrupt.

Source: CSC429 (10).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

a)
State TWO (2) different types of bus.
b)
The size of a word is 32 bits and the size of the block is 256 bits. Determine of the

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following:
i)
The byte for word
ii)
The byte for block,
iii)
Size of the register,
iv)
Size of the data bus.

Source: CSC429 (13).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

Question 5 is based on the following scenario.
"Nurathia Farhana browses the UiTM4fly.com website to book for a return flight from KLIA to LAX. While comparing for a better price, two messages popped up in her screen. Figure 1 (a) and (b) illustrate the conveyed messages.

A hardware failure has been
found. Please restart your
computer now.
(b)

Do you want to be upgraded
to business class now?
(a)

Figure 1: The pop-up messages

a) Which message is the non-maskable interrupt? Justify your answer by giving appropriate reason.
b) Differentiate between maskable and non-maskable interrupt?
c) Outline the common way to handle multiple interrupts as happened in the given scenario.

Source: CSC429 (11).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.

a) Convert 100.5 to IEEE 754 single precision. Give the answer in hexadecimal.

Answer: 100.510 = 0x42C90000 (sign=0, exponent=10000101, fraction=10010010000000000000000)

- *1 mark for sign, exponent and mantissa.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the 32-bit binary grouping.*
- *1 mark for the hexadecimal result.*

b) Convert the IEEE 754 single-precision value 0x3EC00000 to decimal.

Answer: 0x3EC00000 = 0.37510

- *1 mark for the sign bit.*
- *1 mark for the exponent field.*
- *1 mark for the fraction field.*
- *1 mark for applying the bias and significand.*
- *1 mark for the final decimal value.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.
y = A(A + 2B/C-D 2)
Figure 2
Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.
Table 1: Machine Instruction
One-address
Three-address
Zero-address
PUSHp
LOADp
ADD (p,q,r)
POPp
STORp
SUB (p,q,r)
ADD
ADDp
MUL (p.q.r)
SUB
SUBp
DIV(p,q,r)
MUL
MULp
DIV
DIVp

Source: CSC429 (17).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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